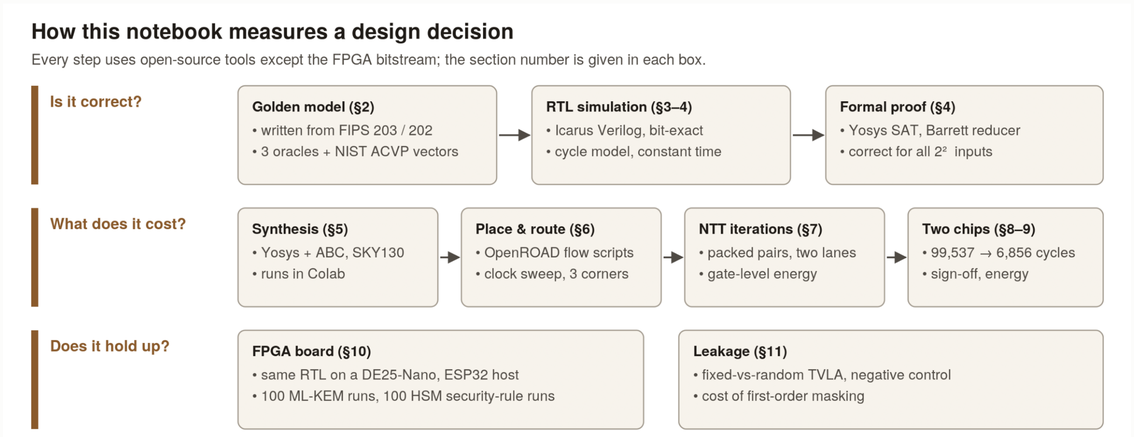
Measuring the Design Decisions of an Open-Source Post-Quantum HSM Chip

ML-KEM-512 NTT and Keccak, from a Python golden model to a SKY130 layout

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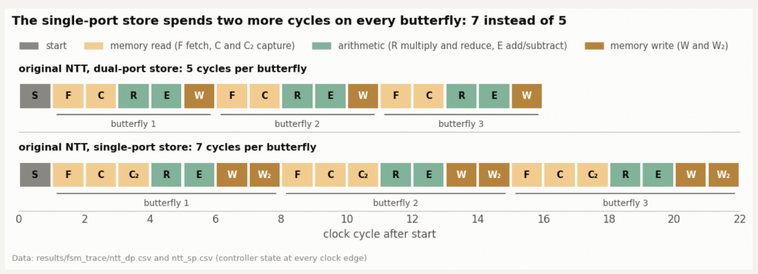
run it in Colab

What did each architectural decision made for the ASIC actually cost in area, latency and energy, what does the design still leak, and can anyone re-derive those numbers with open tools?

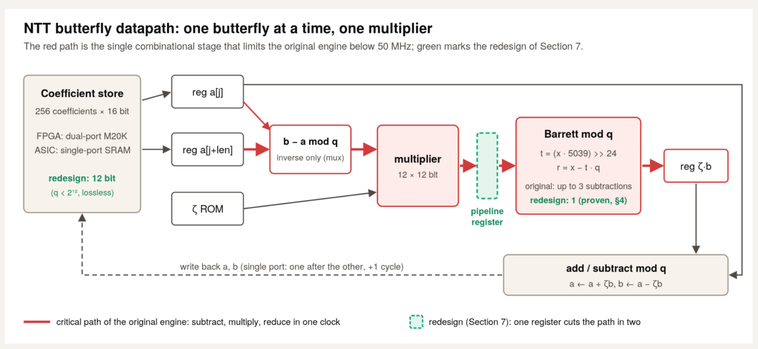


I. Is it correct?

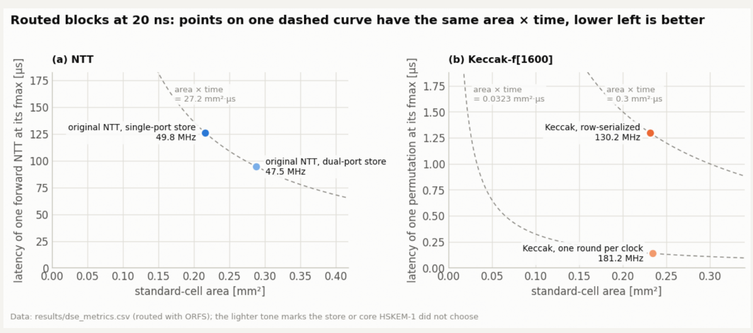
- 25/25 NIST ACVP key generations reproduced byte for byte by the complete RTL
- 0 mismatches over 208,896 NTT coefficients and 404 Keccak permutations, at a constant latency
- The Barrett reducer is proven correct for all 2^{24} inputs, and one subtraction suffices
- The routed netlists reproduce the golden results at gate level



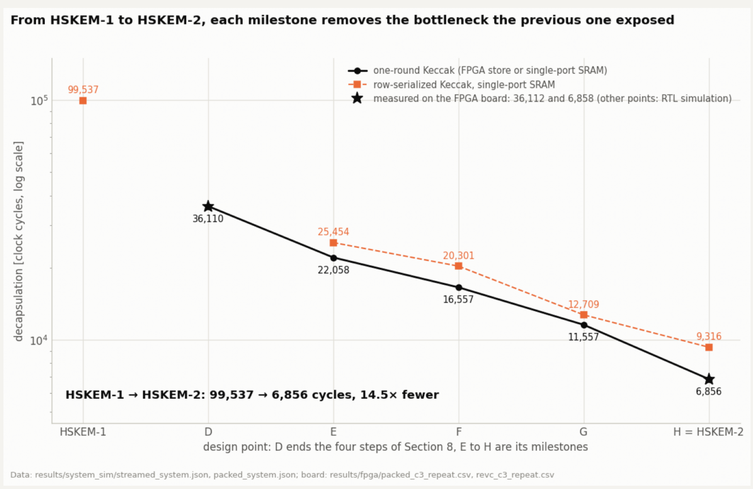
The controller visits the 896 butterflies in the order of FIPS 203, Algorithm 9; the single-port store needs two extra cycles per butterfly.



II. What does it cost?

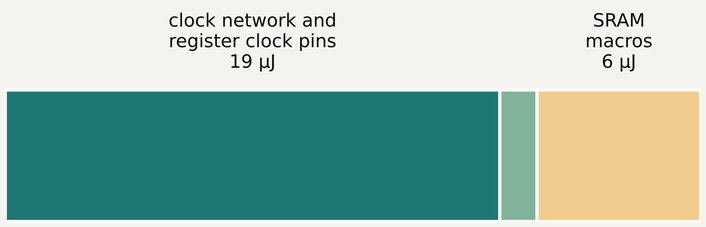


A first measured iteration of the NTT, after routing: fmax 50 → 83 MHz, area -21%, area × time -46%.



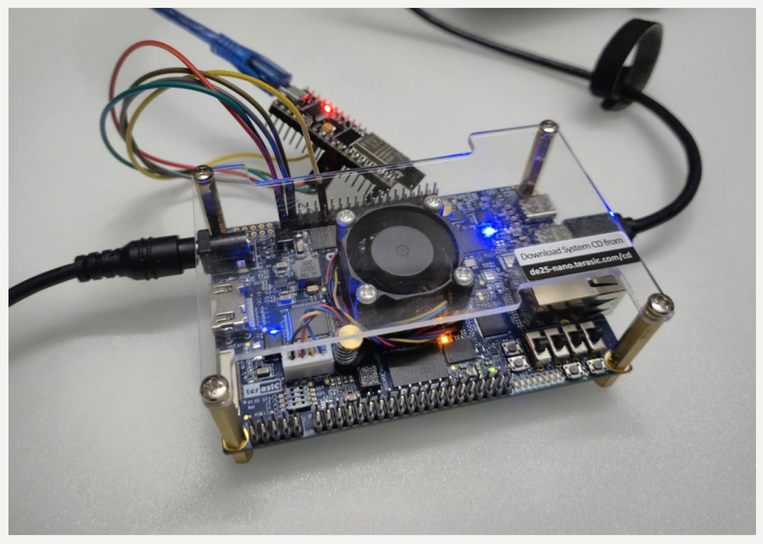
A second iteration stores two coefficients in every SRAM word: a forward NTT takes 14 instead of 130 µs and 0.16 instead of 0.50 µJ. With a two-lane engine and a streamed datapath that removes no check, a decapsulation takes 6,856 instead of 99,537 cycles (HSKEM-1 → HSKEM-2), as the board confirms.

A decapsulation on HSKEM-2: 27 µJ

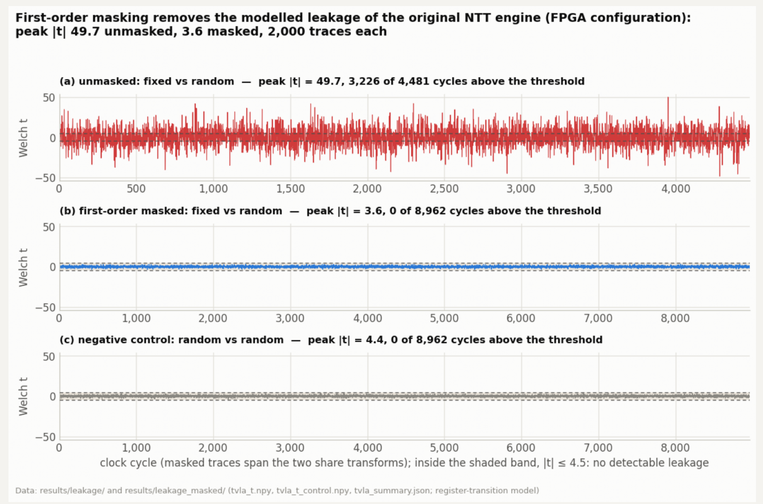


HSKEM-2, the signed-off SKY130 chip of the streamed system, spends 27 µJ per decapsulation instead of HSKEM-1's 0.34 mJ. The rest is set by the integration: gating the clock of idle blocks and the chip selects of the macros would save 32% to 48%.

III. Does it hold up?



The same RTL on the DE25-Nano: 100/100 complete ML-KEM exchanges (both parties on one board) pass.



Simulated TVLA: largest |t| 49.7 unmasked, 3.6 with first-order masking (threshold 4.5).

What transfers

- Weigh a block by how often the system waits for it.
- Measure the integrated chip, not only its blocks.
- Check the check: thirteen results here first looked right and were not.
- Tie the prose to the data: assertions stop the notebook when they part.

Apache-2.0 · open-source tools and the SKY130 PDK throughout (FPGA bitstream: Quartus) · every figure and number is computed from the committed results of the notebook